SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 2- DEBUGGING & OPTIMISATION TASK

Course Code:	CSA1223	Course Title:	Computer Architecture
CO Assessed:	CO2	Assessment:	ASSESSMENT TOOL3- DEBUGGING & OPTIMISATION TASK
Weightage:	20%	Total Marks:	100
CO2 Statement:	*Apply integer and floating-point arithmetic algorithms including Booth's multiplication, restoring/non-restoring division, and IEEE 754 standard operations to solve fixed-point and floating-point computational problems. (BL3)*		
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Department:	CSE (AI/ML)	Date:	12/08/26

76%

ANNEXURE A

1. A program performing signed integer addition using 2's complement produces incorrect results when adding large values (e.g., +120 and +50 in 8-bit representation). Debug the issue by analyzing the binary computation and identifying whether overflow has occurred. Propose a solution to correctly detect and handle overflow in such cases.

2. A processor using a ripple carry adder shows significant delay in executing arithmetic operations in a real-time system. Debug the performance bottleneck by examining how carry propagates through each bit. Suggest an optimized solution using a carry look-ahead adder, and explain how it reduces delay.

3. A multiplication unit implementing Booth's algorithm gives incorrect results for certain negative operands. Debug the step-by-step execution of the algorithm, focusing on bit-pair checking, arithmetic shifts, and sign extension. Identify the possible source of error and suggest corrections to ensure accurate signed multiplication.

4. An embedded system using the restoring division algorithm experiences inefficiency due to repeated restoration steps, leading to increased execution time. Debug the algorithm's workflow and identify where unnecessary operations occur. Propose an optimized approach using the non-restoring division method, explaining how it improves performance.

5. A scientific application using IEEE 754 single precision floating-point representation produces inaccurate results when adding very small and very large numbers. Debug the issue by analyzing exponent alignment, normalization, and rounding errors. Suggest optimization techniques, such as using double precision or algorithmic adjustments, to improve accuracy.

Code of Conduct:

I K. Yoganand (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Yoganand

MARKS ALLOCATION

	Identification of Errors / Bugs (20%)	Debugging Approach & Analysis (20%)	Correctness of Debugged Solution (25%)	Optimization Techniques Applied (20%)	Testing and Documentation (15%)	
Q1	3	4	5	4	2	18
Q2	4	3	4	3	3	17
Q3	3	4	3	2	2	14
Q4	4	3	2	2	1	12
Q5	3	3	2	4	3	15

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RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Identification of Errors / Bugs	20%	Accurately identifies all errors and issues with complete understanding of the problem(4)	Identifies most errors with minor omissions(3)	Identifies limited errors; misses key issues(2)	Unable to identify errors or misunderstands the problem(1)
Debugging Approach & Analysis	20%	Uses effective debugging techniques with clear analysis and root cause identification(4)	Applies suitable debugging methods with minor gaps(3)	Uses basic debugging methods with limited analysis(2)	No systematic debugging approach followed(0)
Correctness of Debugged	25%	Provides completely corrected solution with	Provides working	Partially correct solution with	Incorrect solution or fails to resolve

Solution Process	25%	Logical, systematic steps with correct approach throughout(5)	Mostly correct steps with minor mistakes(4)	Disorganized steps; multiple errors(3)	No clear process(0)
Accuracy of Calculation	20%	All calculations accurate; correct final answer(4)	Minor calculation errors(3)	Frequent errors; incorrect final answer(2)	No valid calculations(0)
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance(3)	Basic interpretation with minor omissions(2)	Limited or unclear interpretation(1)	No interpretation(0)